

**Thesis for obtaining the academic degree
Bachelor of Science**

**Probing Fast Radio Burst Environments:
Cross-Correlating Deep and Shallow
FRB Surveys with KiDS DR5**

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Abstract

The abstract is a short summary of the thesis in English. Together with the German summary, both summaries have to fit on this page.

Kurzfassung

Hier steht eine Kurzfassung der Arbeit in deutscher Sprache inklusive der Zusammenfassung der Ergebnisse. Zusammen mit der englischen Zusammenfassung muss sie auf diese Seite passen.

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1 Introduction

- relevance
- emerging probes $\rightarrow$ different papers cite
- whats plan: frb, galaxy survey $\rightarrow$ combination of both

2 Theoretical Background

2.1 Cosmological Background

The foundation of modern cosmology is based on the *Cosmological Principle*, which states that the Universe is homogeneous and isotropic on large scales. Homogeneity implies that the large-scale properties of the Universe are statistically uniform across different locations, whereas isotropy means that no direction is preferred. Although this principle is introduced as a fundamental assumption in modern cosmology, it also represents a hypothesis that can be tested through observations. Current astronomical evidence strongly supports the validity of the cosmological principle on the largest observable scales [Scrimgeour_2012, Laurent_2016].

Based on the cosmological principle, the Universe can be described by the *Friedmann-Lemaître-Robertson-Walker* (FLRW) metric

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right]$$

with the spatial comoving coordinates (r, θ, ϕ) , the speed of light c , and the cosmic time t [CervantesCota_2011]. The parameter k describes the curvature of the Universe, which can take values of -1 , 0 , or $+1$ for open, flat, and closed geometries, respectively. The scale factor $a(t)$ describes the relative expansion of the Universe and is conventionally normalized to its present value,

$$a(t_0) = 1.$$

For earlier cosmic times, the scale factor satisfies $a(t) < 1$, whereas in an expanding Universe it increases with time and can exceed unity in the future. A static Universe would correspond to $\dot{a}(t) = 0$, resulting in a vanishing Hubble parameter [Huterer_2018].

The expansion rate of the Universe is quantified by the Hubble parameter,

$$H(t) = \frac{\dot{a}(t)}{a(t)},$$

which describes the relative change of the scale factor with cosmic time. At the present time

$$H_0 = H(t_0)$$

is referred to as the Hubble constant [**CervantesCota_2011**].

The expansion of space introduces a distinction between physical distances and comoving distances. While the physical distance between two objects changes with time due to the expansion of the Universe, the comoving distance remains constant for objects that follow the general expansion of space. The physical or so called proper distance is related to the comoving distance χ by

$$D(t) = a(t)\chi.$$

Thus, comoving coordinates remove the effect of the cosmic expansion and provide a convenient description of the large-scale structure of the Universe [**Ishak_2019**].

The expansion of the Universe can be observed through the cosmological redshift. Due to the expansion of space, the wavelength of photons emitted by distant objects is stretched during their propagation. The redshift is defined as

$$z = \frac{\lambda - \lambda_0}{\lambda_0},$$

where λ_0 is the emitted wavelength and λ is the observed wavelength. The redshift is directly related to the scale factor through

$$1 + z = \frac{a(t_0)}{a(t_e)} \tag{2.1}$$

with t_e denoting the time of emission. Consequently, the observed redshift provides information about the expansion history of the Universe and can be used to determine cosmological distances. The line-of-sight comoving distance to an object at redshift z is given by

$$\chi(z) = \chi_H \int_0^z \frac{dz'}{E(a(z'))},$$

where $\chi_H = c/H_0$ is the Hubble distance and $E(a) = H(a)/H_0$ is the dimensionless Hubble parameter [Hogg_2000].

2.2 Friedmann Equations and Cosmic Expansion

Following the cosmological principle, the matter and energy content of the Universe can be described by a perfect-fluid energy-momentum tensor. In combination with the FLRW metric, Einstein's field equations reduce to the *Friedmann equations*

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3}$$

which determine the evolution of the scale factor $a(t)$. Here ρ and p denote the energy density and pressure of the cosmic fluid, Λ is the cosmological constant, and G is the gravitational constant [Huterer_2018].

The critical density at present time is defined as the total energy density for which the Universe is spatially flat ($k = 0$) and is given by,

$$\rho_c = \frac{3H_0^2}{8\pi G}.$$

The ratio between the density of a cosmological component and the critical density defines the corresponding density parameter,

$$\Omega_i = \frac{\rho_i}{\rho_c}.$$

with Ω_m , Ω_r , Ω_Λ , and Ω_k denoting the density parameters of matter, radiation, vacuum (dark) energy, and curvature, respectively.

Consequently, the Friedmann equation can be expressed in terms of the density parameters [BartelmannSchneider_2001]:

$$H^2 = H_0^2 [\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda].$$

Using ??, the dimensionless Hubble parameter can also be written as

$$E(z) = \sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}}.$$

2.2.1 The Λ CDM Model

The current standard model of cosmology is the so-called Λ CDM model. It describes the evolution of the Universe assuming the presence of cold dark matter (CDM) and a cosmological constant Λ , which is associated with dark energy and the accelerated expansion of the Universe. The model is in good agreement with a wide range of cosmological observations [PeeblesRatra_2003]. For the late-time Universe (low redshift) probed by extragalactic sources, the radiation density is negligible and the expansion history can be approximated by the contributions from matter, curvature, and dark energy. Assuming a spatially flat Universe, the curvature term also vanishes, yielding [Gong_2011]

$$E(z) \approx \sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}}.$$

2.3 Fast Radio Bursts

Fast Radio Bursts (FRB) are bright, transient radio pulses with durations of milliseconds or less and peak flux densities ranging from 50 mJy to 100 Jy [Petroff_2019]. They have been detected across a frequency range of 110 MHz [Pleunis_2021] to 8 GHz [Zhang_2018].

The isotropic-equivalent energy distribution of the observed FRB population follows a power-law distribution described by

$$N(E_{\text{iso}}) \propto E_{\text{iso}}^{-1.8},$$

indicating that high-energy bursts are increasingly rare compared to lower-energy events. The negative power-law index of approximately -1.8 describes the decrease in the occurrence rate with increasing isotropic-equivalent energy.

Furthermore, the implied isotropic energies of FRBs are within a few orders of magnitude of those released in gamma-ray bursts and supernova explosions, suggesting that FRBs may originate from highly energetic astrophysical processes [Petroff_2019, Zhang_2021].

Consequently, a broad range of progenitor models has been proposed, including neutron stars, magnetars, white dwarfs, black holes ultra-luminous X-ray sources, active galactic nuclei, and more exotic scenarios such as superconducting cosmic strings [Petroff_2022].

Despite releasing within a single millisecond an amount of energy comparable to what the Sun radiates over three days [Petroff_2019], FRBs are typically located at cosmological distances. As their radio emission propagates across the expanding Universe, it spreads over an immense area, so that the signal received on Earth is roughly a thousand times weaker than that of a mobile phone located on Earth's Moon [SpaceCom_2018] [ZU POPULAERWISSENSCHAFTLICH ???].

The signal from FRBs is dispersed as it propagates through the ionized interstellar and intergalactic medium, causing lower-frequency components to arrive later than higher-frequency components. This makes FRBs appear as a frequency-dependent sweep in time, which can be described by the following relation:

$$DM = \frac{2\pi m_e c}{e^2} \Delta t \left(\frac{1}{\nu_{\text{low}}^2} - \frac{1}{\nu_{\text{high}}^2} \right)^{-1},$$

where ν_{low} and ν_{high} are the lower and higher observing frequencies, Δt is the time delay between the arrival of the two frequency components, c is the speed of light, e is the elementary charge, and m_e is the electron mass. The dispersion measure DM quantifies the integrated column density of free electrons between the source and the observer:

$$DM \propto \int_0^d dl n_e(l),$$

where n_e represents the free electron density and d the distance between the source and the observer [Petroff_2019].

The origin of FRBs is established through their large DM , which exceed the expected contribution from the Galactic interstellar medium along the corresponding line of sight. For a source located within the Milky Way, typical values predicted by Galactic electron-density models are of the order of $DM_{\text{MW}} \lesssim 50 \text{ pc cm}^{-3}$.

In contrast, observed FRBs often exhibit significantly larger values $DM_{\text{FRB}} \gtrsim 1000 \text{ pc cm}^{-3}$ [Reischke_2021].

The first FRB, later named the Lorimer Burst, was identified in archival pulsar survey data acquired in 2001 and published in 2007 [Lorimer_2007].

Initially, only a small number of FRBs had been detected, leading to the assumption that they represented rare, singular events. The situation changed decisively with the commissioning of CHIME in 2018, whose wide field of view drove an explosion in detections and increased the known FRB population by several orders of magnitude. The latest CHIME/FRB catalog contains 4539 detected burst events originating from 3641 unique FRB sources, including 981 bursts from 83 known repeating sources [CHIMEFRB2026]. As shown in ??, the known FRB sources exhibit an approximately isotropic distribution across the sky, consistent with an extragalactic population and suggesting no preferred spatial direction. The shown distribution is based on the FRB sample available up to 2022 and serves as an illustration of the large-scale spatial distribution of FRBs.

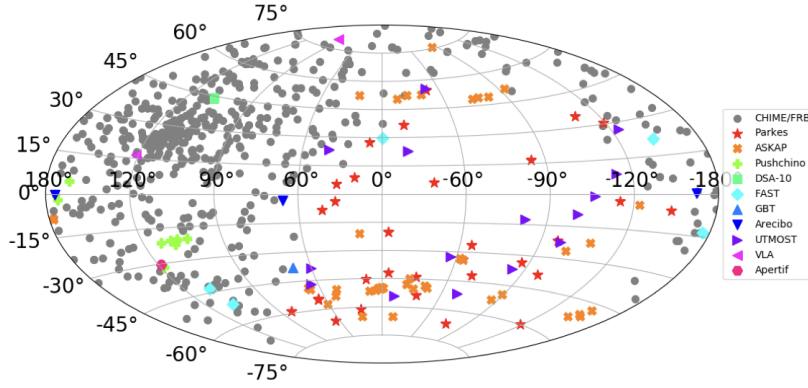


Figure 2.1: Aitoff projection of the known FRB population in Galactic coordinates based on observations available up to 2022. The sample is predominantly composed of CHIME/FRB detections, whose non-uniform sensitivity and sky coverage influence the observed distribution. After correcting for these observational biases, the underlying FRB population is consistent with an isotropic distribution expected for a cosmological source population [Petroff_2022].

2.3.1 Repeating FRBs

While the majority of FRBs have been observed as one-off events, a subset of sources has been found to emit multiple bursts over time. The existence of so-called

repeaters immediately places a fundamental constraint on emission physics: any cataclysmic, single-event progenitor mechanism is directly ruled out for these sources, since the central engine must be capable of sustaining burst activity over timescales of years [Petroff_2019].

Some repeating FRBs also exhibit periodic modulation of their activity windows, as observed in FRB 20180916B [Ng_2023], suggesting either a binary system geometry or precession of the emitting object. Whether all FRBs are intrinsically capable of repeating remains an open question [Petroff_2022].

2.3.2 Progenitor Mechanisms

A broad classification of proposed FRB progenitor models distinguishes between two main categories: catastrophic, one-off events and non-catastrophic scenarios involving long-lived central engines. Catastrophic models can account only for non-repeating FRBs and are therefore incompatible with the observed behavior of repeating sources. In contrast, non-catastrophic models allow for multiple burst episodes over extended periods. Among the proposed progenitors, compact stellar remnants are considered the most promising candidates. In particular, neutron stars and magnetars are strongly supported by both theoretical models and an increasing body of observational evidence [Petroff_2019].

Magnetars are neutron stars distinguished by surface magnetic fields many orders of magnitude stronger than those of ordinary radio pulsars, making them the most strongly magnetised stellar remnants known [Petroff_2019]. They arise primarily through the core collapse of massive stars in supernova explosions and are therefore intrinsically young objects whose formation rate is expected to closely mirror the cosmic star formation history (SFH), preferentially found in or near regions of active star formation [Pellicciari_2023]. The landmark detection in April 2020 of an FRB-like burst from the Galactic magnetar SGR 1935+2154 provided the most direct empirical link between magnetars and FRBs [CHIMEFRB_2020]. Population synthesis work using the CHIME/FRB catalogue finds that the hypothesis of FRBs tracking the SFH is not ruled out by current data, with a small delay preferred, consistent with a scenario in which young magnetars formed through core-collapse supernovae serve as the dominant progenitor class [MengDeng_2025].

Neutron stars formed through delayed channels represent a second important progenitor class, motivated by FRBs localised to environments inconsistent with active star formation. FRBs 180924 and 190523, found in massive, low-star-formation-rate galaxies with large offsets from the host centre, display environments similar to those of short gamma-ray burst hosts associated with mergers of binary neutron-star or

neutron star-black hole binaries [Wang_2020]. The globular cluster localisation of FRB 20200120E reinforces this point, as such an old environment is generally not associated with recent star formation. Isolated core-collapse supernovae occur within approximately 3 Myr to 48 Myr after a burst of star formation, whereas binary interactions can delay some core-collapse events to about 200 Myr [Zapartas_2017]. Accretion-induced collapse (AIC) and compact-object merger channels exhibit much broader delay-time distributions extending to several gigayears [Ruiter_2019]. As a consequence, this progenitor population does not closely track the cosmic SFH but evolves more slowly with redshift [Lin_2022].

These two progenitor classes predict observably distinct redshift distributions [Lin_2022], forming the theoretical basis for the host-population bias modelling adopted in this work, in which each class is assigned a redshift-dependent bias $b(z)$ encoding how strongly its occurrence rate evolves relative to the cosmic star-formation rate. The parameterisation and specific values are detailed in ??.

WAS IST MIT DEN FRAGEN AN KLARA WEGEN DER EDGE CASES SIEHE SLACK?????

2.4 Galaxy Surveys and Catalogs

2.4.1 The Kilo-Degree Survey

The Kilo-Degree Survey (KiDS) is a wide-field optical imaging observation conducted with the 2.6 m VLT Survey Telescope (VST) at ESO’s Paranal Observatory, covering 1347 deg^2 complemented by the VIKING survey to create a nine-band optical-to-near-infrared survey [deJong_2013, Wright_2024]. The data used in this work are drawn from the fifth and final data release (DR5, KiDS-Legacy), which achieves a 5σ limiting magnitude of 24.8 mag in the r -band with a median seeing of 0.7 arcsec [Wright_2024, KiDS_DR5]. Photometric redshifts are calibrated using approximately 126000 spectroscopically confirmed sources [Wright_2025].

2.5 Structure Formation

Within the Λ CDM framework, the large-scale structure of the Universe emerged from the gravitational amplification of small primordial density perturbations. According to the gravitational instability paradigm, quantum fluctuations seeded during an inflationary phase were stretched to cosmological scales and subsequently grew into the filaments, voids, and galaxy clusters observed today [Bernardeau_2002,

Riotto__2003]. The growth of perturbations depends sensitively on the cosmic expansion history, and the overall amplitude of matter fluctuations at the present time is conventionally parameterised by σ_8 [**BartelmannSchneider__2001**].

2.6 Correlation Functions and Angular Power Spectrum

The statistical properties of the large-scale matter distribution are quantified through correlation functions and their Fourier-space counterparts, the power spectra.

The density contrast $\delta(\mathbf{x})$ is defined as the fractional deviation of the local matter density $\rho(\mathbf{x})$ from the mean cosmic density $\bar{\rho}$:

$$\delta(\mathbf{x}) := \frac{\rho(\mathbf{x}) - \bar{\rho}}{\bar{\rho}}.$$

The two-point correlation function is defined as

$$\xi(\mathbf{y}) := \langle \delta(\mathbf{x}) \delta(\mathbf{x} + \mathbf{y}) \rangle,$$

where the angle brackets denote an ensemble average. Statistical homogeneity implies that ξ depends only on the separation $\mathbf{y}$ and not on the absolute position $\mathbf{x}$, while statistical isotropy further implies that it depends only on the magnitude $y = |\mathbf{y}|$.

For a discrete tracer population, the two-point correlation function can be interpreted as describing the excess probability of finding a pair of tracers at a given separation relative to an unclustered random distribution. To describe the Fourier-transform pair of $\delta(\mathbf{x})$ and $\hat{\delta}(\mathbf{k})$, we adopt the convention

$$\hat{\delta}(\mathbf{k}) = \int d^3x \delta(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}; \quad \delta(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \hat{\delta}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}.$$

The three-dimensional matter power spectrum is then defined through

$$\langle \hat{\delta}(\mathbf{k}) \hat{\delta}^*(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} - \mathbf{k}') P(k),$$

where δ_D denotes the Dirac delta function, reflecting statistical homogeneity in Fourier space. The power spectrum depends only on the magnitude $k = |\mathbf{k}|$, implying that it is isotropic in Fourier space [**BartelmannSchneider__2001**].

When the tracers of interest are observed as projected distributions on the celestial sphere, it is common to describe the fields using a spherical-harmonic decomposition. On sufficiently large angular scales, a flat-sky Fourier transform is not generally adequate because of the curvature of the celestial sphere. A scalar field on the sphere can instead be expanded as

$$f(\hat{\mathbf{n}}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}),$$

where $\hat{\mathbf{n}}$ denotes the direction on the sky and $Y_{\ell m}$ are the spherical harmonics. The corresponding harmonic coefficients are given by

$$a_{\ell m} = \int d\Omega f(\hat{\mathbf{n}}) Y_{\ell m}^*(\hat{\mathbf{n}}).$$

The characteristic angular scale associated with a multipole ℓ is approximately ℓ^{-1} [RafieiRavandi_2021]. For a statistically isotropic field, the angular power spectrum is defined through the covariance of the harmonic coefficients,

$$\langle a_{\ell m} a_{\ell' m'}^* \rangle = C_{\ell} \delta_{\ell\ell'} \delta_{mm'}.$$

An unbiased estimator of the angular power spectrum for an ideal full-sky observation is

$$\hat{C}_{\ell} = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2.$$

For a finite or incomplete sky, additional effects such as the survey mask and mode coupling have to be taken into account [Hivon_2002].

The cross-correlation between two distinct projected fields A and B is described by the angular cross-power spectrum,

$$\langle a_{\ell m}^A a_{\ell' m'}^{B*} \rangle = C_{\ell}^{AB} \delta_{\ell\ell'} \delta_{mm'}.$$

A non-zero cross-power spectrum indicates statistical correlation between the two fields on the corresponding angular scales. For projected tracer fields, the angular fields can be written as line-of-sight integrals over the corresponding three-dimensional density contrasts,

$$\begin{aligned} A(\hat{\mathbf{n}}) &= \int d\chi W^A(\chi) \delta^A(\chi \hat{\mathbf{n}}), \\ B(\hat{\mathbf{n}}) &= \int d\chi W^B(\chi) \delta^B(\chi \hat{\mathbf{n}}), \end{aligned} \quad (2.2)$$

where χ denotes the comoving radial distance and $W^A(\chi)$ and $W^B(\chi)$ are the corresponding radial kernel functions. Using the spherical-harmonic expansion together with the Fourier representation of the three-dimensional fields, the full-sky angular cross-power spectrum can be written as

$$C_\ell^{AB} = \frac{2}{\pi} \int_0^\infty k^2 dk \int d\chi W^A(\chi) j_\ell(k\chi) \int d\chi' W^B(\chi') j_\ell(k\chi') P_{AB}(k; \chi, \chi'),$$

where j_ℓ denotes the spherical Bessel function and P_{AB} is the three-dimensional cross-power spectrum of the two tracer fields. The Limber approximation states that for approximately $\ell \gtrsim \mathcal{O}(10)$, the double integral collapses to a single line-of-sight integral [**Limber_1953**]:

$$C_\ell^{AB} \approx \int \frac{d\chi}{\chi^2} W^A(\chi) W^B(\chi) P_{AB} \left(k = \frac{\ell + 1/2}{\chi}, z(\chi) \right). \quad (2.3)$$

The use of $\ell + 1/2$, rather than simply ℓ , provides a more accurate version of the Limber approximation, particularly at moderately large multipoles [**LoVerdeAfshordi_2008**].

The angular cross-power spectrum therefore depends on the radial overlap of the two tracer populations, their relation to the underlying matter density field through their bias factors, and the shape and redshift evolution of the underlying matter power spectrum.

Consequently, the FRB-galaxy angular cross-power spectrum provides a sensitive probe of the spatial relation between FRBs and galaxies and can be used to constrain the redshift distribution, clustering properties, and host-population characteristics of FRBs [**RafieiRavandi_2021, Shirasaki_2017**].

3 Methodology

This chapter describes the analysis pipeline developed in this thesis to model the angular auto- and cross-power spectra of Fast Radio Bursts and photometrically selected galaxies. The pipeline is implemented in Python and follows the theoretical formalism established in ??.

All numerical operations are carried out with NumPy [numpy] and SciPy [scipy]. Background cosmological quantities such as the comoving distance and the Hubble rate are evaluated with the astropy.cosmology module [astropy], while the linear and nonlinear matter power spectra as well as the linear growth factor are obtained from the hmf package [hmf]. All figures are produced with Matplotlib [matplotlib].

HIER NOCHMAL KUERZEN JE NACHDEM WIE LANG ES AM ENDE IST!

3.1 Fiducial Cosmology and Numerical Grids

Throughout this thesis a spatially flat Λ CDM cosmology as introduced in ?? is assumed. The fiducial parameter values are taken from the Planck 2018 baseline analysis of the combined TT,TE,EE+lowE+lensing likelihood [Planck2018] and are summarised in ??.

Table 3.1: Fiducial cosmological parameters assumed throughout this thesis. The values correspond to the Planck 2018 baseline TT,TE,EE+lowE+lensing results [Planck2018].

Parameter	Symbol	Value
Hubble constant	H_0 [km/(s Mpc)]	67.36
Reduced Hubble parameter	h	0.6736
Total matter density	$\Omega_{m,0}$	0.3153
Baryon density	$\Omega_{b,0}$	0.022 37
Fluctuation amplitude	σ_8	0.8111
Scalar spectral index	n_s	0.9649
CMB temperature	$T_{\text{CMB},0}$ [K]	2.7255

MAYBE REDUCED HUBBLE PARAMETER RAUS!!

The corresponding background model is realised as a `FlatLambdaCDM` instance of `astropy.cosmology` [`astropy`], from which the line-of-sight comoving distance $\chi(z)$ and the Hubble rate $H(z)$ are evaluated numerically following the relations given in ??.

All redshift-dependent quantities are evaluated on a common logarithmically spaced grid of 1000 points covering $z \in [0.01, 5]$. The logarithmic spacing provides a finer sampling at low redshift, where the radial weight functions of the galaxy sample have substantial support. This ensures a higher numerical resolution in the redshift range that contributes significantly to the calculation. Angular power spectra are evaluated on a dense grid of integer multipoles $\ell \in [10, 1000]$. The lower limit is set by the validity range of the Limber approximation discussed in ??, which requires $\ell \gtrsim \mathcal{O}(10)$ [Limber_1953, LoVerdeAfshordi_2008], while the upper limit restricts the analysis to angular scales for which the adopted nonlinear matter power-spectrum prescription remains sufficiently reliable.

BIS HIER GEPRUEFT AUSSER DIE FETTEN SACHEN OBEN: POTENTIELL DIE GENAUE RECHNUNG MIT DEN L LOGARITHMISCH USW RAUSNEHMEN

3.2 Nonlinear Matter Power Spectrum

The three-dimensional matter power spectrum $P(k, z)$ introduced in ?? is computed using the `hmf` package [`hmf`].

To obtain a continuous representation of the resulting values, a bicubic spline interpolation is performed using the `RectBivariateSpline` routine from `SciPy` [`scipy`].

The resulting interpolator is shown in ??.

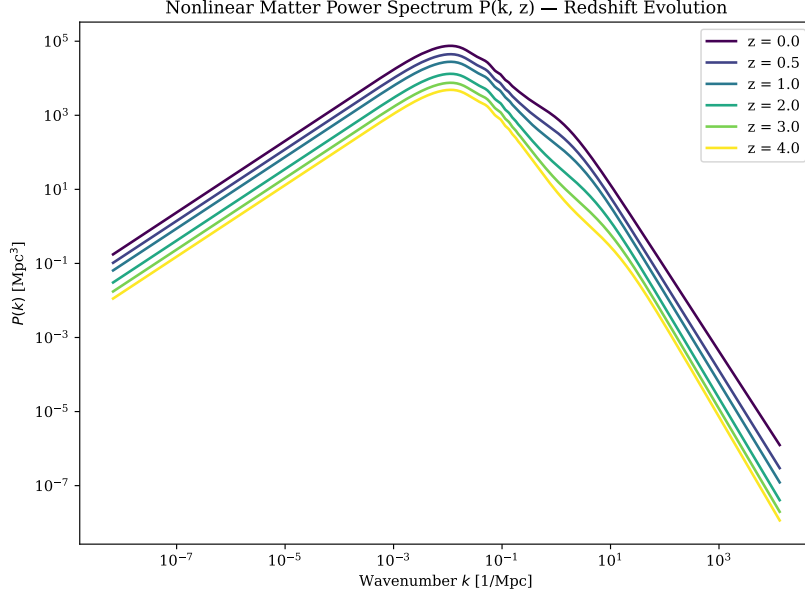


Figure 3.1: Nonlinear matter power spectrum $P(k, z)$ evaluated from the bicubic spline interpolator at six representative redshifts.

3.3 Redshift Distributions

3.3.1 FRB Redshift Distribution

Since the FRB population considered in this thesis is not associated with individually measured redshifts, its radial selection function has to be modelled. Following Reischke, Hagstotz and Lilow [Reischke_2021], the normalised FRB redshift distribution is assumed to take the form

$$n(z) \propto z^2 \exp(-\alpha z), \quad (3.1)$$

where the parameter α controls the steepness of the high-redshift tail and thereby the effective depth of the survey. The z^2 prefactor reflects the growth of the comoving volume element at low redshift, whereas the exponential cutoff encodes the decreasing detectability of bursts at large distances.

Two survey configurations are considered, both taken from [Reischke_2021] and summarised in ???. A shallow survey is characterised by $\alpha = 3.5$ and a total of

$N_{\text{tot}} = 5 \times 10^3$ detected bursts, while a deep survey is described by $\alpha = 2.0$ and $N_{\text{tot}} = 5 \times 10^4$ bursts. Both configurations are assumed to cover a sky fraction of $f_{\text{sky}} = 0.9$, consistent with the near-all-sky coverage achievable by wide-field radio interferometers such as CHIME (??).

Table 3.2: FRB survey configurations adopted in this thesis, following [Reischke_2021]. The steepness parameter α enters the redshift distribution of ??, while the total number of bursts N_{tot} and the sky fraction f_{sky} determine the shot-noise level according to ??.

Configuration	α	N_{tot}	f_{sky}
Shallow	3.5	5000	0.9
Deep	2.0	50 000	0.9

MAYBE HERE DEN ABSCHNITT OBEN EIN BISCHEN KUERZEN

The distribution of ?? is normalised numerically on the redshift grid defined in ??, such that $\int n(z) dz = 1$. Both distributions are shown in ??.

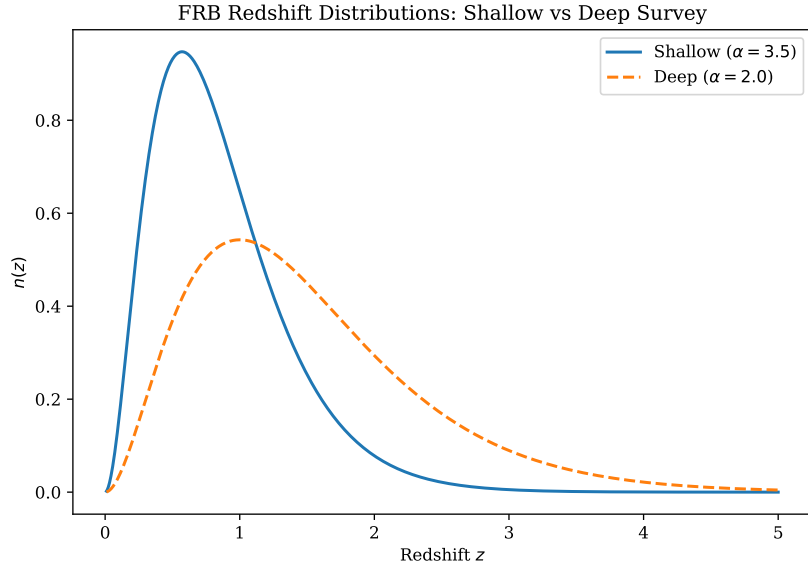


Figure 3.2: Normalised FRB redshift distributions of ?? for the shallow ($\alpha = 3.5$) and deep ($\alpha = 2.0$) survey configuration. The shallow configuration peaks near $z \approx 0.6$ and falls off rapidly beyond $z \approx 1.5$, whereas the deep configuration peaks near $z \approx 1.0$ and retains appreciable support out to $z \approx 3$. The deep configuration therefore overlaps substantially more with the high-redshift galaxy bins, which is the decisive factor for the cross-correlation signal discussed in ??.

The corresponding comoving distance distribution has to be transformed accordingly. Conservation of the number of sources, $n(\chi) d\chi = n(z) dz$, yields

$$n(\chi) = n(z) \frac{dz}{d\chi}, \quad (3.2)$$

with the Jacobian $dz/d\chi = H(z)/c$ following from the definition of the comoving distance in ??.

3.3.2 Galaxy Redshift Distribution

In contrast to the FRB sample, the galaxy redshift distribution is not modelled analytically but taken from the fifth and final data release of the Kilo-Degree Survey (KiDS-Legacy) introduced in ??.

In order to combine the galaxy distributions with the FRB distributions, the tabulated $n_i(z)$ are linearly interpolated onto the common redshift grid of ??. Outside the tabulated range the distributions are set to zero, reflecting the absence of calibrated sources beyond the support of the published data. The resulting distributions are shown in ??.

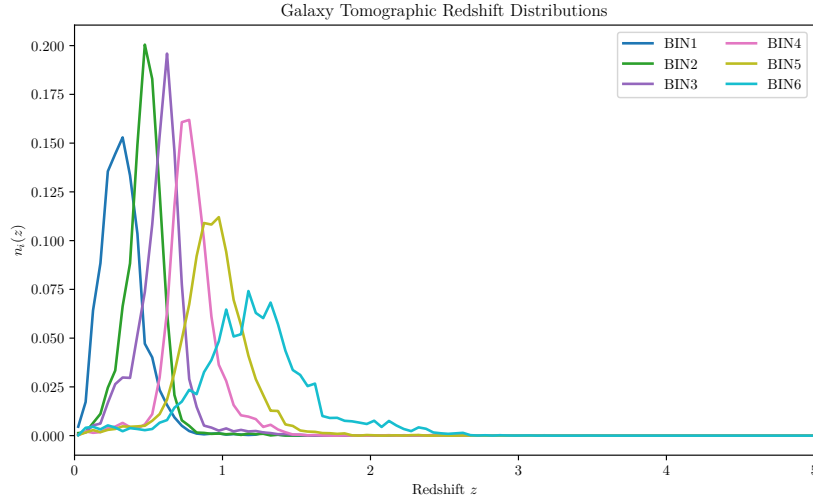


Figure 3.3: Tomographic redshift distributions $n_i(z)$ of the six KiDS-Legacy galaxy bins as published in [Wright_2025].

3.4 Tracer Bias Models

3.4.1 FRB Host-Population Bias

As motivated in ??, the two dominant progenitor classes are expected to trace the underlying matter distribution differently. This difference is encoded in a redshift-dependent linear bias of the power-law form

$$b(z) = b_0 (1 + z)^\delta, \quad (3.3)$$

where b_0 denotes the bias amplitude at $z = 0$ and the exponent δ parameterises its redshift evolution. The fiducial values adopted for the two progenitor classes are listed in ?. The magnetar channel is assigned a low amplitude but a steep evolution, consistent with a young population whose host haloes become progressively rarer, and therefore more strongly biased, towards higher redshift. The delayed neutron-star channel is assigned a higher amplitude but a much shallower evolution, reflecting a population that resides in massive hosts already at low redshift and whose clustering is largely decoupled from the instantaneous star-formation rate.

Table 3.3: Fiducial parameters of the FRB host-population bias model of ? for the two progenitor classes considered in this thesis [`TODO_frb_bias`].

Progenitor class	b_0	δ
Magnetars	1.0	0.8
Neutron stars	1.5	0.2

HIER NOCH QUELLE FINDEN

3.4.2 Galaxy Bias

For the galaxy sample a constant-clustering bias model is adopted, in which the comoving clustering amplitude of the sample is assumed to remain constant over the redshift range spanned by each tomographic bin. Under this assumption the bias evolves inversely with the linear growth factor,

$$b_g^i = \frac{0.95}{D_+(\langle z \rangle_i)}, \quad (3.4)$$

which is the standard prescription used in photometric survey forecasts [Johnston_2025].
BIS HIER GECHECKT

Here $D_+(z)$ denotes the linear growth factor normalised to $D_+(0) = 1$, evaluated with `hmf` [hmf], and $\langle z \rangle_i$ is the mean redshift of the i -th tomographic bin. The latter is computed from the tabulated distributions as the discrete first moment

$$\langle z \rangle_i = \frac{\sum_j z_j n_i(z_j)}{\sum_j n_i(z_j)}.$$

Since ?? is evaluated at the mean redshift of the bin rather than at every sampling point, each tomographic bin is assigned a single constant bias value. This is a deliberate simplification: the bias is a slowly varying function across the width of an individual bin, so that replacing it by its value at the effective redshift introduces an error that is small compared to the uncertainty of the bias model itself. The resulting mean redshifts and bias values are listed in ??. AUS DEM PART HIER OBEN KOENNT MAN ZUMINDEST DIE FORMEL WEGNEHMEN UND DAS DARUNTER EIGENTLICH AUCH

Table 3.4: Properties of the six KiDS-Legacy tomographic galaxy bins. The mean redshift $\langle z \rangle_i$ is computed from the published redshift distributions and the bias b_g^i follows from ??.

Bin	$\langle z \rangle_i$	b_g^i
1	0.335	1.135
2	0.477	1.221
3	0.587	1.291
4	0.789	1.423
5	0.940	1.525
6	1.224	1.721

3.5 Radial Weight Functions

The radial kernels $W^A(\chi)$ and $W^B(\chi)$ introduced in ?? weight the three-dimensional density contrast along the line of sight. For a discrete tracer population whose fluctuations are related to the matter density contrast by a linear bias, the kernel is given by the product of the normalised radial selection function and the bias of the population.

For the FRB sample the weight function is therefore constructed on the redshift grid as

$$W_{\text{FRB}}(z) = b(z) n(z), \quad (3.5)$$

with $n(z)$ from ?? and $b(z)$ from ??.

For the galaxy sample the corresponding kernel of the i -th tomographic bin reads

$$W_g^i(z) = b_g^i n_i(z). \quad (3.6)$$

Both kernels are subsequently transformed to comoving distance using the Jacobian introduced in ??:

$$W(\chi) = W(z), \frac{H(z)}{c}, \quad (3.7)$$

which gives the radial kernels entering the projection integral.

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3.6 Angular Power Spectrum Computation

The angular power spectra are computed from the Limber-approximated projection integral of ??.

For each multipole the integrand

$$\frac{W^A(\chi) W^B(\chi)}{\chi^2} P\left(k = \frac{\ell + 1/2}{\chi}, z(\chi)\right)$$

is evaluated on the comoving-distance grid corresponding to the redshift grid of ?? and integrated with the composite trapezoidal rule.

3.6.1 Auto-Correlation Analysis

For the auto-correlation of a single tracer population, the two kernels in ?? coincide, $W^A = W^B$, and the projection integral reduces to a weighted line-of-sight average of the matter power spectrum. Auto-spectra are computed for both FRB survey configurations and both progenitor classes, as well as for each of the six tomographic galaxy bins.

In contrast to the cross-spectra discussed below, auto-spectra receive an additive Poissonian shot-noise contribution from the discreteness of the tracer field, which is treated in ?. ? illustrates the relative importance of signal and noise for a representative FRB configuration, while ? compares the signal-only spectra of the two survey configurations for both progenitor classes. The galaxy auto-spectra of all six tomographic bins are shown in ?.

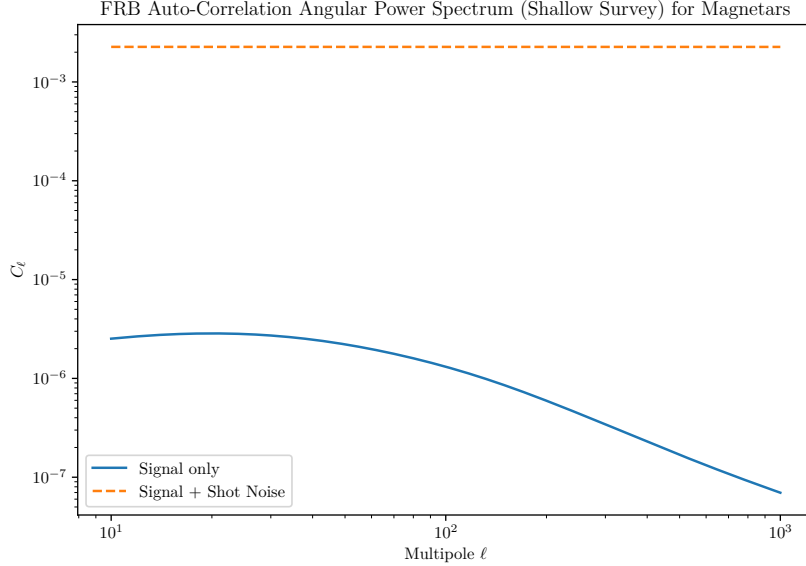


Figure 3.4: FRB auto-correlation angular power spectrum for the magnetar population in the shallow survey configuration, shown with and without the Poissonian shot-noise contribution. The clustering signal peaks at $\ell \approx 20$ and falls off steeply thereafter, whereas the shot noise is scale-independent and exceeds the signal by roughly three orders of magnitude across the entire multipole range. For a burst sample of only 5×10^3 events the FRB auto-correlation is therefore completely noise-dominated, which is the central motivation for combining it with a dense galaxy sample in a cross-correlation analysis.

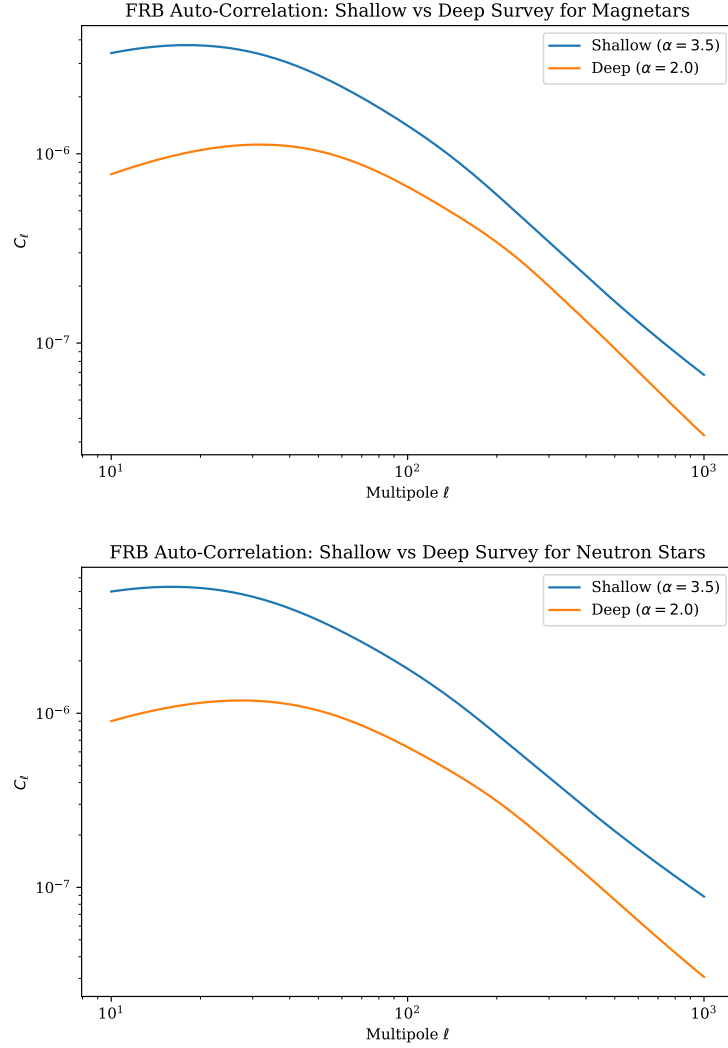


Figure 3.5: Signal-only FRB auto-correlation angular power spectra for the shallow and deep survey configuration, shown for the magnetar population (top) and the delayed neutron-star population (bottom). In both cases the shallow configuration yields the larger amplitude, since its narrower redshift distribution projects a smaller radial range onto the same angular scale and therefore suffers less from line-of-sight dilution. Comparing the two panels shows that the choice of progenitor model changes the overall amplitude but leaves the shape of the spectrum largely unaffected, which illustrates the degeneracy between bias amplitude and redshift evolution in the auto-spectrum alone. HIER NOCHMAL SCHAUEN OB WIRKLICH DAS GANZE SO RICHTIG ERKLAERT IST

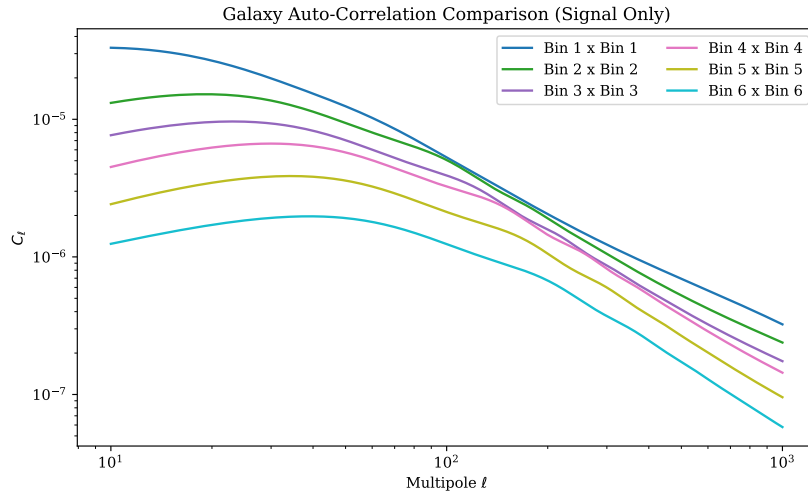


Figure 3.6: Signal-only galaxy auto-correlation angular power spectra of the six KiDS-Legacy tomographic bins. The amplitude decreases systematically from the low-redshift to the high-redshift bins: although the bias increases with redshift according to ??, this is more than compensated by the broader radial extent of the high-redshift bins and by the smaller amplitude of the matter power spectrum at earlier times. In addition, the peak of the spectrum shifts towards higher multipoles for the higher bins, since a fixed comoving scale subtends a smaller angle when viewed from a larger distance. HIER AUCH NOCHMAL REINGUCKEN UND VERSTEHEN FALLS NOETIG

3.6.2 Cross-Correlation Analysis

The angular cross-power spectrum between the FRB sample and the i -th tomographic galaxy bin is obtained from ?? by inserting the two distinct kernels of ?? and ?. Cross-spectra are computed for all combinations of two progenitor classes, two survey configurations and six tomographic bins, yielding $2 \times 2 \times 6 = 24$ spectra in total.

Regarding shot-noise, FRBs reside in galaxies and a subset of their hosts may be contained in the KiDS sample, which would introduce a small residual Poisson term. Given the vast difference in number density between the two samples, this contribution is neglected here.

The resulting spectra are presented in three complementary ways. ?? compares the six tomographic bins for a fixed progenitor class and survey configuration, ?? contrasts the two progenitor classes, and ?? contrasts the two survey configurations.

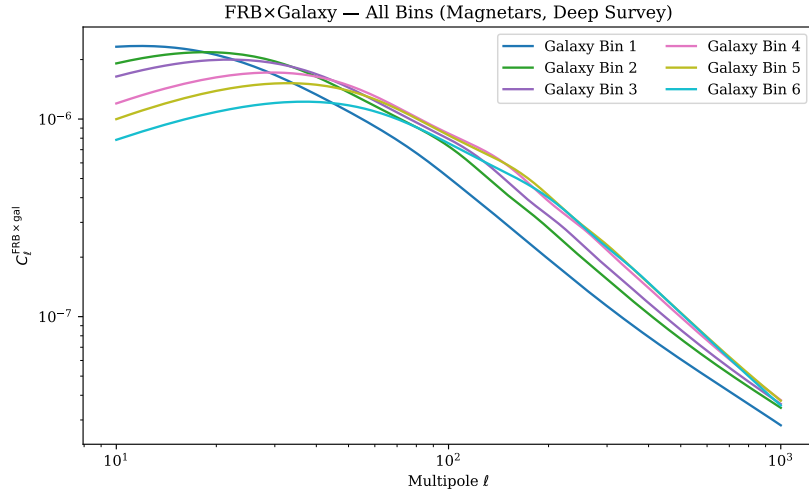


Figure 3.7: FRB×galaxy angular cross-power spectra for the magnetar population in the deep survey configuration, shown separately for the six tomographic galaxy bins. The amplitude decreases monotonically from the first to the sixth bin, directly reflecting the decreasing radial overlap between the FRB redshift distribution of ?? and the progressively higher-redshift galaxy bins of ?. The cross-spectrum therefore acts as a tomographic probe of the FRB redshift distribution.

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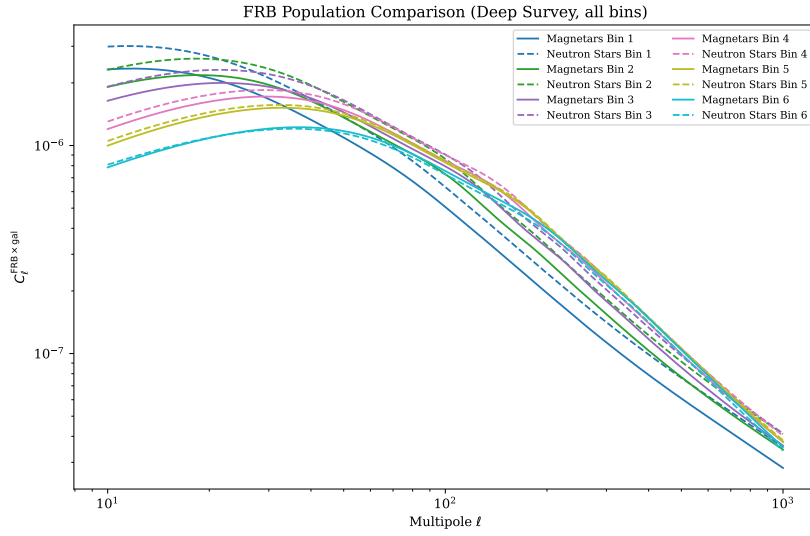


Figure 3.8: Comparison of the FRB×galaxy cross-power spectra of the magnetar and the delayed neutron-star population in the deep survey configuration, shown for all six tomographic galaxy bins. In the sixth bin, whose mean redshift lies above the crossing point of the two bias models, the two predictions become nearly indistinguishable. The separation between the progenitor classes is thus bin-dependent, which is what makes a tomographic cross-correlation analysis informative.

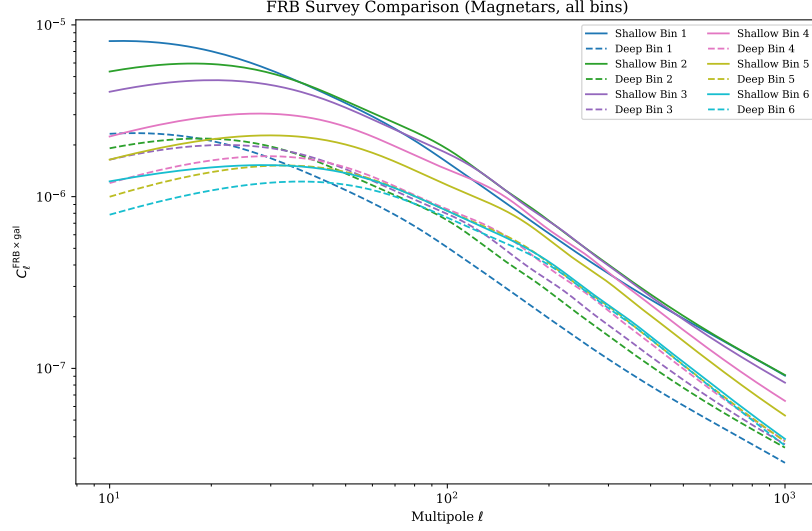


Figure 3.9: Comparison of the FRB×galaxy cross-power spectra of the shallow and deep survey configuration for the magnetar population, shown for all six tomographic galaxy bins.

3.7 Shot Noise and Survey Coverage

Because both tracers are discrete point processes, the measured auto-spectra contain an additive Poissonian noise term. For a sample of mean projected number density $\bar{n}$ per steradian, this contribution is scale-independent and given by

$$N_\ell = \frac{1}{\bar{n}}. \quad (3.8)$$

For the FRB samples the number density follows from the total number of detected bursts and the observed sky fraction listed in ??,

$$\bar{n}_{\text{FRB}} = \frac{N_{\text{tot}}}{4\pi f_{\text{sky}}}.$$

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Finally, the incomplete sky coverage of a realistic survey is accounted for only through the sky fraction f_{sky} . A measurement on a masked sky does not yield the true

angular power spectrum but a mode-coupled pseudo-spectrum, whose deconvolution requires the full mode-coupling matrix of the survey mask [**Hivon_2002**]. As this thesis is concerned with the theoretical prediction of the angular power spectra rather than with their estimation from observed maps, no mask is applied and no mode-coupling correction is performed; the effect of partial sky coverage enters solely through the reduced number of independent modes per multipole.

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4 Results

4.1 Fisher Matrix

5 Conclusion & Outlook

A Appendix

This appendix can contain supplementary material such as code listings, drawings or other extended material that does not fit into the main text. In most cases, all relevant results should be presented in the main part of the thesis and an appendix is optional.

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